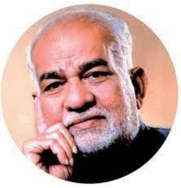
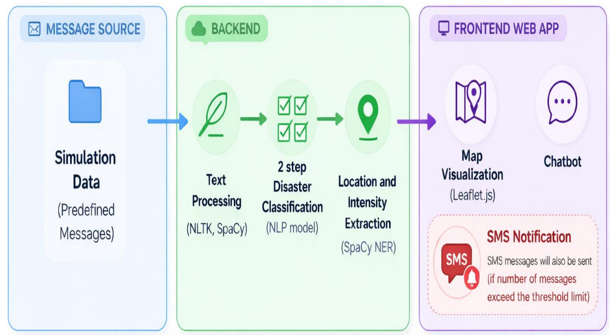


BVRIT HYDERABAD
College of Engineering for Women
(UGC Autonomous)
Department of Computer Science & Engineering

5 Rakshak AI – Intelligent Disaster Response System

ABSTRACT
Rakshak AI is a disaster detection system that analyzes text messages using the all-MiniLM-L6-v2 model. It identifies whether a message indicates a disaster and extracts type, location, severity, and confidence score. The system also provides map visualization, alerts, chatbot support, and safe location suggestions to improve disaster awareness and response.

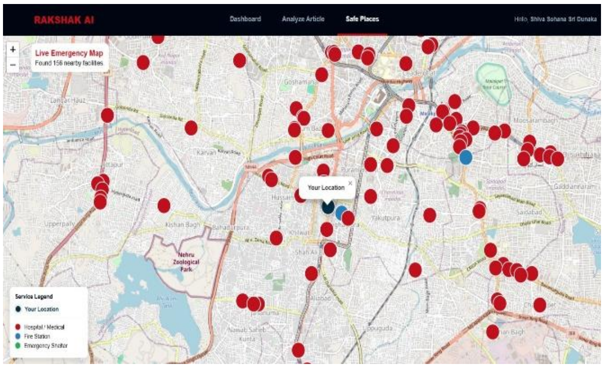
Architecture Diagram



UNIQUENESS
- Two stage disaster detection
- Map based visualization system
- Threshold based alerts
- Safe places finder

METHODOLOGY
The system uses MiniLM embeddings and machine learning models. It first classifies messages as disaster or non-disaster. If disaster is detected, it predicts type and severity. Location is extracted using spaCy NER. This data is simultaneously plotted on a map using different color codes. When a threshold number of messages are received from a particular area, SMS alerts will be sent to users in that area.

Results and Analysis



Safe places visualization

SOCIETAL USE
Rakshak AI helps users by providing early disaster alerts, safety guidance, and nearby emergency locations. It improves preparedness and supports quick response during disasters.

SDG 13 - Climate Action

REFERENCES
[1] He, C., & Hu, D. (2025). Social media analytics for disaster response: Classification and geospatial visualization framework. *Applied Sciences*, *15*(8), 4330.

[2] Khallouli, W., Li, J., Huang, J., Rabadi, G., & Kovacic, S. (2025). An integrated machine learning approach for identifying emergency rescue messages on social media during natural disasters. *Social Network Analysis and Mining*, *15*(1), 50.

GITHUB LINK
https://github.com/sohanasri1806/rakshak-ai



23WH1A05H4 **K.Manya Sree**

23WH1A05H7 **D.Shiva Sohanasri**

23WH1A05H9 **G.Rishika**

Faculty Mentor : Ms. D.Swapna **Email Id: swapna.d@bvrithyderabad.edu.in**